



दिल्ली प्रौद्योगिकी विश्वविद्यालय
DELHI TECHNOLOGICAL UNIVERSITY
(Formerly Delhi College of Engineering)
(Estd. By Govt. of NCT of Delhi vide Act 6 of 2009)



Master of Technology (M.Tech)

Discipline: Artificial Intelligence (AI)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(w.e.f. 2023-24)

DELHI TECHNOLOGICAL UNIVERSITY
(Formerly Delhi College of Engineering)
Shahbad Daultapur, Main Bawana Road, Delhi-42
(Academic-PG)

M.Tech Artificial Intelligence						
Scheme for Full Time M. Tech.						
SEMESTER I						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI501	Advanced Data Structure and Algorithm	Core	4	3-0-2	24	500-599*
AI505	AI and Machine Learning	Core	4	3-0-2		
AI507	Natural Language Processing	Core	4	3-1-0		
AI509	Artificial Neural Networks	Core	4	3-1-0		
AI531	Elective 1	Departmental Elective 1	4	3-1-0/3-0-2/2-0-4/0-0-8		
AI525		Self-Study	2	-		
AI523		Skill Enhancement Course 1	2	-		
UEC501		Audit Course	0	0-0-2		
SEMESTER II						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level

AI502	Evolutionary Computing	Core	4	3-1-0	24	500-599*
AI504	Pattern Recognition	Core	4	3-0-2		
AI532	Elective 2	Departmental Elective 2	4	3-1-0/3-0-2/2-0-4/0-0-8		
AI534	Elective 3	Departmental Elective 3	4	3-1-0/3-0-2/2-0-4/0-0-8		
UCC502		Research Methodology and IPR	4	3-1-0/3-0-2/2-0-4/0-0-8		
AI546/548		Skill Enhancement Course 2/Industrial Training	4	-		
		NHEQF Level				6.5
SEMESTER III						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI601	Advanced System and Network Security	Core	4	3-0-2	16	600-699*
UEC601		Open Elective 1	4	3-1-0/3-0-2/2-0-4/0-0-8		
AI603		Minor Project/ Research Thesis/Patent	8	-		
SEMESTER IV						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level

AI602		Major Project/Research Thesis/Patent	16	-	16	-
		NHEQF Level				7.0

M.Tech Artificial Intelligence						
Scheme for Part Time M. Tech.						
SEMESTER I						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI501	Advanced Data Structure and Algorithm	Core	4	3-0-2	12	500-599*
AI505	AI and Machine Learning	Core	4	3-0-2		
AI507	Natural Language Processing	Core	4	3-1-0		
SEMESTER II						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI509	Artificial Neural Networks	Core	4	3-1-0	12	500-599*
AI502	Evolutionary Computing	Core	4	3-1-0		
AI532	Elective 2	Departmental Elective 2	4	3-1-0/3-0-2/2-0-4/0-0-8		

		NHEQF Level				6.5
SEMESTER III						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI504	Pattern Recognition	Core	4	3-0-2	12	500-599*
AI531	Elective 1	Departmental Elective 1	4	3-1-0/3-0-2/2-0-0-4/0-0-8		
AI525		Self-Study	2	-		
AI523		Skill Enhancement Course 1	2	-		
UEC501		Audit Course	0	0-0-2		
SEMESTER IV						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI534	Elective 3	Departmental Elective 3	4	3-1-0/3-0-2/2-0-0-4/0-0-8	12	500-599*
UCC502		Research Methodology and IPR	4	3-1-0/3-0-2/2-0-0-4/0-0-8		
AI546/548		Skill Enhancement Course 2/Industrial Training	4	-		
		NHEQF Level				
						6.5

SEMESTER V						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI601	Advanced System and Network Security	Core	4	3-0-2	16	600-699*
UEC601		Open Elective 1	4	3-1-0/3-0-2/2-0-4/0-0-8		
AI603		Minor Project/Research Thesis/Patent	8	-		
SEMESTER VI						
Course Code	Course Name	Type	Cr	L-T-P	Total Credits	Level
AI602	16	Major Project/Research Thesis/Patent	16	-	16	
		NHEQF Level				7.0

List of Departmental Electives

Elective 1	
AI531A	Programming language for AI
AI531B	Advance Database Management System
AI531C	Digital Image Analysis
AI531D	Statistics for Data Analytics
AI531E	Information Theory & Coding
AI531F	Optimization Techniques
AI531G	Probability & Statistical Theory
AI531H	Applications of AI & its impact on Society
AI531I	Soft Computing
AI531J	Advance Computer Graphics
AI531K	Robotics Engineering
Elective 2	
AI532A	Speech and Biometric Processing
AI532B	Data Mining and Data Warehousing
AI532C	Deep Learning
AI532D	Computer Vision
AI532E	Semantic Web
AI532F	Security and Privacy in Social Networks
AI532G	Mobile and Cellular Network Security
AI532H	Game Theory
AI532I	Intelligent Agents
AI532J	Genetic Algorithm & Evolutionary Algorithm
Elective 3	

AI534A	Augmented Reality and Virtual Reality
AI534B	Big Data Analytics
AI534C	Digital Forensics
AI534D	Medical AI
AI534E	Data Visualization
AI534F	Statistical Machine Learning
AI534G	Block Chain and Applications
AI534H	Quantum AI
AI534I	Intellectual Property Rights
AI534J	Internet of Things, Applications and Security
AI534K	Business Data Analytics
AI534L	Predictive Analytics and Statistics

Note: A student must earn at least 08 credits from 04 credit elective courses and at least 06 credits from 03 credit elective courses from a basket of electives specific to that programme. The remaining credits may be earned from electives offered for other programmes including Massive Open Online Course's (MOOC). In case of MOOC's only 02 credits per MOOC course will be considered.

Course Evaluation

Evaluation in each course is based on the weights assigned to the various components of the course curriculum. These components are designated as under.

CWS	Class work sessional
MTE	Mid-term examination
ETE	End term Examination
PRS	Practical sessional
PRE	Practical Examination

Total credits	L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	3	0	2	Yes	Yes	15	20	40	25	0
4	3	1	0	Yes	No	25	25	50	0	0
3	3	0	0	Yes	No	20	30	50	-	-
2	2	0	0	Yes	No	20	30	50	-	-
24	0	0	24	No	Yes	0	0	60	40	0
12	0	0	12	No	Yes	0	0	100	-	0

AI501: Advanced Data Structures and Algorithm

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI501	Advanced Data Structures and Algorithm	Core	4	3	0	2	Yes	Yes	15	20	40	25	0

Details of Course

This course covers topics of

Review of Elementary data structures- Binary Trees, Binary Heap, Sorting & Searching Technique

Sparse matrices- Properties of Sparse Matrices, Linked List representation of Sparse Matrices, Analysing Algorithm, Hashing, Universal Hashing, Perfect Hashing

Advanced Data Structures: data structures for combinatorial Optimization – Binomial Heap, Fibonacci Heaps, Red -Black Trees, Augmenting Red – Black Trees to Dynamic order Statics and Interval Applications. Operations on Disjoint Sets –find union problem, Implementing Sets, Dictionaries, Self- Adjusting Trees, Skip lists.

Divide and Conquer approach- Application of Divide and Conquer- Finding Maximum and Minimum, Finding K-th smallest element order statistics, Finding K-th Smallest element, Merge Sort, Randomized Quick Sort.

Graph Algorithms: Definitions for Graph, Algorithms for connectedness, Finding All Spanning Trees in a Weighted Graph and Planarity Testing, Breadth First and Depth First Search, Articulation Point, Cut Edge, Topological Sort, Strongly Connected Components and Single Source Shortest Path and All pair shortest path algorithms, Planer graphs

Greedy Method and Dynamic Programming: General method, knapsack problem, Single Source Shortest path, Job Sequencing with deadline, Scheduling problems. Dynamic Programming: General method, 0/1 knapsack problem, All Pair Shortest Path.

Backtracking: Sum of subsets, 8-queens problem, and Hamiltonian cycles, Traveling Salesman Problem.

Textbooks
1. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, “Introduction to Algorithms”, MIT Press. 2. Aho, Hopcraft & Ulman, The Design and Analysis of Computer algorithms”, Addison Wesley. 3. S. Dasgupta, C. H. Papadimitriou, and U . V . V azirani” Algorithms “ 4. Tannenbaum, “Data Structures”, PHI
Reference Books
1. R.E. Tarjan, “Data Structures and Network algorithms”, SIAM Regional Conference series in applied mathematics.. 2. Ellis Horowitz, Sartaj Sahni Sanguthevar Rajasekaran, “The Design and Analysis of Computer Algorithms”. 3. Rajeev Motwani and Prabhakar Raghavan, “Randomized Algorithms”, Cambridge University Press. 4. Dexter C. Kozan, “The Design & Analysis of Algorithms”, Springer-Verlag. 5. Narsingh Deo-Graph Theory with Application to Engineering and Computer Science,Prentice Hall of India.

AI505 AI and Machine Learning

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI505	AI and Machine Learning	Core	4	3	0	2	Yes	Yes	15	20	40	25	0

Course Outcomes: On completion of the course, the students will be able to

1. Explain the Fundamentals of Artificial Intelligence.
2. Design and Analyze Intelligent Agents
3. Apply Search Algorithms for Problem-Solving
4. Implement Machine Learning Models for Classification and Regression
5. Evaluate and Optimize Machine Learning Model
6. Develop Solutions Using Advanced AI Technique

Pre-requisite: Linear algebra, Theory of Probability, Calculus

Details of Course:

Unit	Contents	Contact Hours
I.	Definition of AI, Scope of AI, History and milestones, AI applications (robotics, NLP, vision), Challenges in AI, Ethics and impact of AI, AI problem-solving techniques (search, reasoning, learning, decision-making), Definition of agents, Percepts and actions, Goals and environments, Agent architectures (reflex, goal-based, utility-based, learning), Agent-environment interactions, Fully vs. partially observable, Deterministic vs. stochastic, Episodic vs. sequential, Agent performance measures, PEAS description, Definition of rationality, Rational action, Human vs. machine rationality, Autonomy and adaptability, Optimality vs. bounded rationality, Resource limitations, Knowledge limitations, State space representation, Defining states and actions, Transitions and goals, Search problem formulation, Pathfinding, Puzzles and planning, Search trees, Search graphs, Properties of search algorithms (completeness, optimality, time complexity, space complexity), Uninformed search: Best first search, informed search: Greedy Best-First Search, A* Search algorithm.	8
II.	Introduction, application and Types of machine learning: Supervised, unsupervised, and reinforcement learning, labeled data, training vs. testing data, applications. Feature selection techniques (Filter methods, Wrapper methods, Embedded methods), scaling and normalization (Min-Max scaling, Z-score standardization), handling missing data (imputation methods: mean, median, KNN-based imputation), Differences between classification and regression, tasks and applications, curve fitting. curse of dimensionality, Curve fitting: Polynomial regression and its limitations. Loss functions: Mean Squared Error (MSE), Cross-entropy loss, Hinge	8

	loss. Train/test split, importance of separate data for training and testing. Cross-validation techniques: K-fold cross-validation, Leave-One-Out Cross-Validation (LOOCV). Bias-variance trade-off: Impact on model performance. Introduction to validation curves and learning curves. Confusion matrix, evaluation metrics: Accuracy, Precision, Recall, Specificity, F1 score, ROC curve, and AUC.	
III.	Introduction to Linear Regression, application examples, Simple Linear Regression, Multiple Linear Regression, Mathematical Foundation: equation of a line, plane, hyperplane (simple and multiple regression), Derivative of cost function, gradient descent algorithm, Introduction to Logistic Regression, role in classification, comparison with linear regression, Applications, logistic function, logits, learning parameters, cost function for logistic regression, gradient descent algorithm in logistic regression, Decision Trees: structure, entropy, information gain, Gini impurity, early stopping, tree pruning, advantages and limitations, use cases. Introduction to Artificial Neural Networks, Single-layer perceptron, Linear separability, MLP architecture, Feedforward networks, Multiple hidden layers, Forward propagation, Backpropagation and Weight updates.	8
IV.	Game playing in AI, Two-player zero-sum games, Game tree representation, Minimax algorithm, Decision-making in adversarial settings, Alpha-beta pruning, Pruning efficiency, Complex game strategies, Expectimax algorithm, Stochastic games, Decision-making under uncertainty, Markov chains basics, State transitions, Hidden Markov Models (HMMs), Markov Decision Processes (MDPs), States, actions, transition models, Rewards in MDPs, Value functions (V(s)), Action value functions (Q(s,a)), Policies, Bellman equations, Optimal policies, V* and Q*, Policy evaluation, Iterative value calculation, Policy iteration (evaluation and improvement), Value iteration, Convergence to optimal values and policies, Applications of MDPs (robot navigation, resource allocation)	9
V.	Introduction to Reinforcement Learning, Introduction to Q-Learning and Temporal Difference (TD) learning. Sarsa algorithm: On-policy learning, Exploration vs. exploitation trade-off. Discount factor and its implications on immediate vs. long-term rewards. Policy iteration: Policy evaluation and policy improvement. Value iteration: Using the Bellman optimality equation for finding optimal policies, TD Learning: Advantages of TD(0) over Monte Carlo methods. Introduction to Actor-Critic methods: Architecture and advantages.	9
	TOTAL	42

Suggested Books

S.No.	Name of Books / Authors/ Publishers
1.	Bishop, C. M. (2006). Pattern Recognition and Machine Learning (1st ed.). Springer.
2.	Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep Learning (1st ed.). MIT Press.
3.	Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly Media.
4.	Sutton, R. S., & Barto, A. G. (2018). Reinforcement Learning: An Introduction (2nd ed.). MIT Press.
5.	James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). Introduction to Statistical Learning (2nd ed.). Springer.
6.	Hastie, T., Tibshirani, R., & Friedman, J. (2017). The Elements of Statistical Learning (2nd ed.). Springer.
7.	Barber, D. (2012). Bayesian Reasoning and Machine Learning (1st ed.). Cambridge University Press.
8.	Chollet, F. (2021). Deep Learning with Python (2nd ed.). Manning Publications.

9.	Murphy, K. P. (2012). Machine Learning: A Probabilistic Perspective (1st ed.). MIT Press.
10.	Russell, S., & Norvig, P. (2021). Artificial Intelligence: A Modern Approach (4th ed.). Pearson.

AI507 Natural Language Processing

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI507	Natural Language Processing	Core	4	3	1	0	Yes	No	25	25	50	0	0

Details of Course

Introduction to NLP and Text Processing

Definition and scope of NLP, history and evolution of NLP, applications and real-world examples, course logistics and expectations, text normalization techniques, tokenization methods, stop words removal, stemming and lemmatization, handling punctuation and special characters.

Linguistic Fundamentals and Language Modeling

Syntax and grammar, sentence structure, part-of-speech (POS) tagging, morphology and word formation, semantics and meaning representation, semantic roles and relations, introduction to language models, n-gram models and applications, smoothing techniques (Laplace, Good-Turing), perplexity as an evaluation metric.

Statistical Methods in NLP

Statistical methods in NLP, Hidden Markov Models (HMMs), Viterbi algorithm for decoding, machine learning approaches to POS tagging, Maximum Entropy Models, Conditional Random Fields (CRFs), evaluation metrics (accuracy, precision, recall).

Parsing and Semantic Analysis

Constituency parsing, context-free grammars (CFGs), parse trees, dependency parsing, dependency relations and structures, parsing algorithms (CKY, Earley), lexical semantics, word sense disambiguation, compositional semantics, semantic role labeling, ontologies and semantic networks.

Distributional Semantics, Word Embeddings, Text Classification, and Sequence Models

Vector space models of semantics, word embeddings (Word2Vec, CBOW, Skip-Gram), GloVe embeddings, evaluating semantic similarity, text classification tasks and applications, feature extraction methods (Bag-of-Words, TF-IDF), traditional machine learning algorithms for classification (Naive Bayes, SVMs), evaluation metrics (precision, recall, F1-score), introduction to neural networks in NLP, recurrent neural networks (RNNs), long short-term memory (LSTM) networks, gated recurrent units (GRUs).

NLP Applications and Ethical Considerations

Sentiment analysis and opinion mining, lexicon-based approaches, machine learning methods without deep learning, dealing with negation and intensifiers, applications in social media and customer feedback, information extraction, named entity recognition (NER), relation extraction, rule-based and statistical methods, evaluation of information extraction systems, text summarization, extractive versus abstractive summarization, extractive summarization techniques (frequency-based methods, TextRank), evaluation of summaries, bias in language resources (gender, racial, cultural biases), ethical considerations in data collection and model deployment, privacy concerns and data protection, guidelines for responsible NLP development, student project presentations, feedback and discussion, review of key concepts, future directions in NLP.

Recommended Textbooks and Resources
1. Jurafsky, D., & Martin, J. H. (2020). <i>Speech and Language Processing</i> (3rd ed.).
2. Bird, S., Klein, E., & Loper, E. (2009). <i>Natural Language Processing with Python</i> . O'Reilly Media.
3. Manning, C. D., Raghavan, P., & Schütze, H. (2008). <i>Introduction to Information Retrieval</i> . Cambridge University Press.
4. Goldberg, Y. (2017). <i>Neural Network Methods for Natural Language Processing</i> . Morgan & Claypool Publishers.

AI509 Artificial Neural Networks

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI509	Artificial Neural Networks	Core	4	3	1	0	Yes	No	25	25	50	0	0

Introduction: A Neural Network, Human Brain, Models of a Neuron, Neural Networks viewed as Directed Graphs, Network Architectures, Knowledge Representation, Artificial Intelligence and Neural Networks Learning Process: Error Correction Learning, Memory Based Learning, Hebbian Learning, Competitive, Boltzmann Learning, Credit Assignment Problem, Memory, Adaption, Statistical Nature of the Learning Process. **Single Layer Perceptron:** Adaptive Filtering Problem, Unconstrained Organization Techniques, Linear Least Square Filters, Least Mean Square Algorithm, Learning Curves, Learning Rate Annealing Techniques, Perceptron –Convergence Theorem, Relation Between Perceptron and Bayes Classifier for a Gaussian Environment Multilayer Perceptron: Back Propagation Algorithm XOR Problem, Heuristics, Output Representation and Decision Rule, Computer Experiment, Feature Detection. Back Propagation: Back Propagation and Differentiation, Hessian Matrix, Generalization, Cross Validation, Network Pruning Techniques, Virtues, and Limitations of Back Propagation Learning, Accelerated Convergence, Supervised Learning.

Textbooks

1. “Artificial Neural Networks”, B. Yegnanarayana, Prentice Hall of India P Ltd 2005.
2. “Neural Networks in Computer Intelligence”, Li Min Fu, TMH 2003.
3. “Neural Networks”, James A Freeman and David M Skapura, Pearson Education 2004.
4. “Introduction to Artificial Neural Systems” Jacek M. Zurada, JAICO Publishing House Ed. 2006.

AI502 Evolutionary Computing

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI502	Evolutionary Computing	Core	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Evolutionary Computation: Biological and artificial evolution, Evolutionary computation and AI, Different historical branches of EC, e.g., GAs, EP, ES, GP, etc., A simple evolutionary algorithm, **Search Operators:** Recombination/Crossover for strings (e.g., binary strings), e.g., one-point, multi-point, and uniform crossover operators, Mutation for strings, e.g., bit-flipping, Recombination/Crossover and mutation rates, Recombination for real-valued representations, e.g., discrete and intermediate recombination, Mutation for real-valued representations, e.g., Gaussian and Cauchy mutations, self-adaptive mutations, etc., Why and how a recombination or mutation operator works.

Selection Schemes: Fitness proportional selection and fitness scaling, Ranking, including linear, power, exponential and other ranking methods, Tournament selection, Selection procedure and its impact on evolutionary search, Search Operators and Representations: Mixing different search operators, An anomaly of self-adaptive mutations, The importance of representation, e.g., binary vs. Gray coding, Adaptive representations.

Evolutionary Combinatorial Optimization: Evolutionary algorithms for TSPs, Evolutionary algorithms for lecture room assignment, Hybrid Evolutionary and local search algorithms

Co-evolution: Cooperative co-evolution, Competitive co-evolution, Niching and Speciation: Fitness sharing (explicit and implicit), Crowding and mating restriction, Constraint Handling: Common techniques, e.g., penalty methods, repair methods, etc.

Genetic Programming: Trees as individuals, Major steps of genetic programming, e.g., functional and terminal sets, initialization, crossover, mutation, fitness evaluation, etc., Search operators on trees. Automatically defined functions, Issues in genetic programming, e.g., bloat, scalability, etc., Multi objective Evolutionary Optimization: Pareto optimality, Multi objective evolutionary algorithms

Suggested books :

S.No.	Name of Books/Authors/Publishers
1	Handbook on Evolutionary Computation, T. Baeck, D. B. Fogel, and Z. Michalewicz, IOP Press, 1997.
2	Genetic Algorithms + Data Structures = Evolution Programs, Third Edition, Z Michalewicz, Springer-Verlag, Berlin, 1996.
3	Evolutionary Computation: Theory and Applications, X. Yao, World Scientific Publ. Co., Singapore, 1999. (ISBN 3-540-65907-2).

AI504 Pattern Recognition

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI504	Pattern Recognition	Core	4	3	0	2	Yes	Yes	15	20	40	25	0

Details of course

Introduction to Pattern Recognition, Feature Detection, Classification, Review of Probability Theory, Conditional Probability and Bayes Rule, Random Vectors, Expectation, Correlation, Covariance, Review of Linear Algebra, Linear Transformations.

Decision Theory, ROC Curves, Likelihood Ratio Test, Linear and Quadratic Discriminants, Fisher Discriminant, Sufficient Statistics, Coping with Missing or Noisy Features. Template-based Recognition, Feature Extraction, Eigenvector and Multilinear Analysis, Training Methods, Maximum Likelihood and Bayesian Parameter Estimation, Linear Discriminant/Perceptron Learning, Optimization by Gradient Descent.

Support Vector Machines, K-Nearest-Neighbor Classification, Non-parametric Classification, Density Estimation, Parzen Estimation. Unsupervised Learning, Clustering, Vector Quantization, K-means, Mixture Modelling, Expectation-Maximization. Hidden Markov Models, Viterbi Algorithm, Baum-Welch Algorithm, Linear Dynamical Systems, Kalman Filtering, Bayesian Networks.

Text Books:	
1.	R. O. Duda, P. Hart, D. Stork, “Pattern Classification”, Second Edition, Wiley, 2000.
2.	C.M. Bishop, “Pattern Recognition and Machine Learning’. Springer, 2007.
3.	C.M. Bishop, “Neural Networks for Pattern Recognition”, Oxford University Press, 1995.
4.	S. Theodoridis and K. Koutroumbas, “Pattern Recognition”, Fourth Edition. Academic Press, 2008.
5.	T. Hastie, R. Tibshirani and J. Friedman, “The Elements of Statistical Learning”, Springer, 2009.

AI601 Advanced Systems and Network Security

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
3	AI601	Advanced Systems and Network Security	Core	4	3	0	2	Yes	Yes	15	20	40	25	0

Prerequisite: Computer Networks

Details of Course

Introduction to Data Security: General Security Principles, Communication Security, Design of Secure System. classical encryption techniques substitution ciphers and transposition ciphers, cryptanalysis, stereography, stream and block ciphers, Intruders, Viruses and related threads.

Conventional Encryption Principles: Conventional encryption algorithms, cipher block modes of operation, location of encryption devices, key distribution Approaches of Message Authentication, Secure Hash Function and HMAC.

Modern Block Ciphers: Block ciphers principals, Shannon's theory of confusion and diffusion, fiestal structure, data encryption standard (DES), strength of DES, cryptanalysis of DES, block cipher modes of operations, triple DES, IDEA encryption and decryption, strength of IDEA, key distribution.

Public-key cryptography Principles: Recent Public key cryptography algorithms such as RSA, DSS etc., digital signatures, digital Certificates, Certificate Authority and key management Kerberos, X.509 Directory Authentication Service.

Latest Trends and solutions in Information Security: Web Security, Firewall Design Principles, Trusted Systems, Intrusion Detection System. Recent trends in cryptosystem.

IP Security: Architecture, Authentication header, Encapsulating security payloads, combining security associations, key management.

Web Security: Secure Socket Layer (SSL) and transport layer security, TSP, Secure Electronic Transaction (SET), Electronic money, WAP security, firewall design principals, Virtual Private Network (VPN) security.

Textbooks
1. “Cryptography and Network Security”, William Stallings, Third Edition, Prentice Hall International. 2. “Basic Methods of Cryptography”, Jan C A, Cambridge University Press. 3. “Information Security Intelligence: Cryptographic Principles & Applications”, Thomas Calabrese, Thomson Learning.
Reference Books
1. “Modern Cryptography: Theory and Practice”, Wenbo Mao, Pearson Education. 2. “Cryptography and Data Security”, Dorothy Elizabeth Robling Denning, Addison Wesley.

AI531A Programming language for AI

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI531A	Programming language for AI	Elective	4	3	0	2	Yes	Yes	15	20	40	25	0

DATA, EXPRESSIONS, STATEMENTS: Python interpreter and interactive mode; values and types: int, float, boolean, string, and list; variables, expressions, statements, tuple assignment, precedence of operators, comments; modules and functions, function definition and use, flow of execution, parameters and arguments. CONTROL FLOW, FUNCTIONS Conditionals: Boolean values and operators, conditional (if), alternative (if-else), chained conditional (if, if-else); Iteration: state, while, for, break, continue, pass; Fruitful functions: return values, parameters, local and global scope, function composition, recursion; Strings: string slices. DICTIONARIES: Operations and methods; advanced list processing – list comprehension Object Oriented Programming: Classes and objects-inheritance-polymorphism. Exception Handling & File Handling: Overview of exception classes and types: try, except. Finally, File processing: reading and Writing files.

Text Books:

1. Reema Thereja, “Python Programming using Problem Solving Approach”, Oxford Publication, 2019.
2. John Zelle, “Python Programming: An Introduction to Computer Science”, Third Edition, Franklin, Beedle & Associates.

AI531B: Advanced Database Management Systems

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI531B	Advanced Database Management	Elective	4	3	0	2	Yes	Yes	15	20	40	25	0

Details of Course

This course covers topics of Relational Databases Integrity Constraints, Extended ER diagram, Relational Algebra & Calculus, Functional, Multivalued and Join Dependency, Normal Forms, Rules about functional dependencies.

Advanced Transaction Processing; Nested and multilevel transaction ,Compensating Transaction & Saga, Long Duration Transaction ,Weak Level of Consistency, Transaction Work flows ,Transaction Processing Monitors, Schedules ,Serializability- Conflict & View

Query Processing ; General Strategy for Query Processing ,Transformations ,Expected size ,Statistics in Estimation ,Query improvement ,View Processing ,Query Processor

Query Optimization: Indexing and Query Optimization, Limitations of Relational Data Model ,Null Values & Partial Information

Parallel and Distributed Databases: Distributed Data Storage – Fragmentation & Replication, Location and Fragment Transparency, Distributed Query Processing and Optimization, Distributed Transaction Modelling & Concurrency Control, Distributed Deadlock ,Commit Protocol, Design of Parallel Databases ,Parallel Query Evaluation

Active Database and Real Time Databases: Triggers in SQL, Event Constraint and Action-ECA rules ,Query Processing & Concurrency Control ,Compensation & Databases Recovery

Textbooks

5. Elmars, Navathe, Somayajulu, Gupta, “Fundamentals of Database Systems”, 4th Edition, Pearson Education, 2007.
6. Garcia, Ullman, Widom, “Database Systems, The complete book”, Pearson

Reference Books

6. Date, Kannan, Swaminathan, “An Introduction to Database Systems”, 8th Edition Pearson Education, 2007.
7. Silberschatz, Korth, Sudarshan, “Database System Concepts”, Mcgraw Hill, 6th Edition, 2006

AI531C Digital Image Analysis

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI531C	Digital Image Analysis	Elective	4	3	1	0	Yes	No	25	25	50	0	0

The origins of Digital Image Processing. Examples of Fields that Use Digital Image Processing, Fundamentals Steps in Image Processing, Elements of Digital Image Processing Systems, Image Sampling and Quantization, Some basic relationships like Neighbors, Connectivity, Distance Measures between pixels, Linear and Non Linear Operations. **Image Enhancement in the Spatial Domain.** Some basic Gray Level Transformations, Histogram Processing, Enhancement Using Arithmetic and Logic operations, Basics of Spatial Filters, Smoothing and Sharpening Spatial Filters, Combining Spatial Enhancement Methods. **Image Enhancement in the Frequency Domain.** Introduction to Fourier Transform and the frequency Domain, Smoothing and Sharpening Frequency Domain Filters, Homomorphic Filtering. **Image Restoration.** A model of The Image Degradation / Restoration Process, Noise Models, Restoration in the presence of Noise Only Spatial Filtering, Periodic Noise Reduction by Frequency Domain Filtering, Linear Position-Invariant Degradations, Estimation of Degradation Function, Inverse filtering, Wiener filtering, Constrained Least Square Filtering, Geometric Mean Filter, Geometric Transformations. **Image Compression.** Coding, Interpixel and Psychovisual Redundancy, Image Compression models, Elements of Information Theory, Error free comparison, Lossy compression, Image compression standards. **Image Segmentation.** Detection of Discontinuities, Edge linking and boundary detection, Thresholding, Region Oriented Segmentation, Motion based segmentation. Representation, Boundary Descriptors, Regional Descriptors, Use of Principal Components for Description, Introduction to Morphology, Some basic Morphological Algorithms.

Text Books:

1. Jayaraman," Digital image processing", Tata Mcgraw Hill, 2011.
2. R.C. Gonzalez and R.E. Woods, "Digital Image Processing", Third Edition, 2007.

AI531D Statistics for Data Analytics

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI531D	Statistics for Data Analytics	Elective	4	3	1	0	Yes	No	25	25	50	0	0

UNIT-I

Statistical Methods: Definition and scope of Statistics, concepts of statistical population and sample. Data: quantitative and qualitative, attributes, variables, scales of measurement nominal, ordinal, interval and ratio. Presentation: tabular and graphical, including histogram and ogives, consistency and independence of data with special reference to attributes.

UNIT-II

Measures of Central Tendency: mathematical and positional. Measures of Dispersion: range, quartile deviation, mean deviation, standard deviation, coefficient of variation, Moments, absolute moments, factorial moments, skewness and kurtosis, Sheppard's corrections.

UNIT-III

Bivariate data: Definition, scatter diagram, simple, partial and multiple correlation (3 variables only), rank correlation. Simple linear regression, principle of least squares and fitting of polynomials and exponential curves.

UNIT-IV

Index Numbers: Definition, construction of index numbers and problems thereof for weighted and unweighted index numbers including Laspeyre's, Paasche's, Edgeworth- Marshall and Fisher's Ideal Index numbers. Errors in Index numbers. Chain index numbers, conversion of fixed based to chain based index numbers and vice-versa. Consumer price index numbers. Uses and limitations of index numbers.

TEXT BOOKS:

1. Goon A.M., Gupta M.K. and Dasgupta B. (2002): Fundamentals of Statistics, Vol. I

& II, 8th Edn. The World Press, Kolkata.
2. Gupta, S. C. and Kapoor, V.K. (2008): Fundamentals Of Mathematical Statistics,
4thEdition (Reprint), Sultan Chand & Sons

AI531E Information Theory and Coding

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI531E	Information Theory and Coding	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Probability: Basic terms and concept, Random variables, Probability density and distribution function, Marginal and conditional distributions. Expectation. Markov's inequality, Chebyshev inequality, Binomial, Poisson and Gaussian distributions, Introduction to Markov process. Uncertainty and Information, Shannon Entropy, Joint and conditional Entropies. Mutual Information, Uniquely decipherable and Instantaneous codes, Noiseless coding problem. Source coding Theorem, Block coding, construction of optimal codes, Huffman's & Shannon – Fano methods. Discrete memory less channel, channel capacity. BSC and other channels. Information measure for continuous ensembles capacity of AWGN channel. Error control coding: The channel coding Theorem, Application to BSC, Source Coding with fidelity criteria. Types of codes, error and error control strategies, Linear block codes, syndrome and error detection, Minimum distance, Error detecting and correcting capabilities of a block code, Syndrome decoding, Hamming codes. Cyclic codes, Generator and parity: check matrices, encoding, syndrome computation and error detection and decoding. BCH codes, decoding of the BCH codes. Introduction to RS codes. Convolution codes, Maximum likelihood decoding, Viterbi algorithm, Turbo codes.

Text Books:

1. R Ash, "Information Theory", Dover Science Publications.
2. Cover and Thomas, "Element of Information Theory", John Wiley & Sons.

Reference Books:

1. Shulin and Daniel J. Costello Jr, "Error Control coding: Fundamental & Application", Prentice Hall, Inc.
2. Simon Haykin, "Communication Systems", Wiley Student Edition.
3. H.C.Taneja, "Statistical methods for Engineering and Sciences", I.K. International.

AI531F Optimization Techniques

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI531F	Optimization Techniques	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Basics of Optimization Techniques: Historical Development, Engineering application of Optimization, Formulation of design problems as mathematical programming problems, classification of optimization problems, Constrained and Un-constrained Optimization. Linear Programming Problem: Convex Sets, Hyper plane, Graphical method, Simplex method, Revised simplex method, Duality in linear programming (LP), Sensitivity analysis, other algorithms for solving LP problems, Transportation, assignment and other applications. Non-Linear Programming Problem: Quadratic Forms, Convex Non-linear Programming Problem, Method of Lagrange multipliers, Kuhn–Tucker Theory, Convex Quadratic Programming Problem, Separable Programming, Geometric Programming, Polynomial Programming Problem. Dynamic Programming: Introduction, Forward and Backward dynamic Programming, Recursive Relations, Search Techniques, Uniform, Sequential, and Fibonacci search Techniques, Univariate search methods, Steepest Descent method, Conjugate Directions method, Fletcher Reev Methods. Queuing Systems: Introduction, Characteristics of Queuing Models, Models for arrival and service time, Kendall’s notation for representing Queuing Models, Birth and Death Processes Advanced Techniques of Optimization: Introduction, Particle Swarm Optimization, Bacteria Foraging, Ant Colony Optimization, and Genetic algorithms for optimization and search.

Text Books:

1. S. S. Rao, “Engineering optimization: Theory and practice”, New Age International (P) Limited, Third Edition, 1998.
2. J. C. Pant, “Introduction to Optimization Operations Research”, Jain Bros, Seventh Edition, 2008.
3. K.V. Mital and C. Mohan, “Optimization Methods in Operations Research and systems Analysis”, New Age International (P) Limited, Publishers, Third Edition, 1996.

Reference Books:

1. H.A. Taha, “Operations Research : An Introduction” , PHI Pvt. Ltd.

2. Kwang Y.Lee and El-Sharkawi, “Modern Heuristic Optimisation Techniques”, Wiley Publication.

AI531G Probability and Statistical Theory

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
3	AI531G	Probability and Statistical Theory	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Probability theory, through set and Relative Frequency: Experiments and Sample Spaces, Discrete and Continuous Sample Spaces, Events, Probability and Axioms, Probability as a Relative Frequency, Joint and Conditional Probability, Random Variables, Distribution Function, Density Function, Operations on Random Variables, Sampling Distributions, Estimation, Testing of Hypotheses, Neyman-Pearson Fundamental Lemma, Chi-square goodness of fit test and its applications.

Text Books:

1. P. Peebles, "Probability random variables and random signal principles", Fourth Edition, McGraw Hill, 2013.
2. A. Papoulis, S. Pillai, "Probability: Random Variables and Stochastic Processes and Probability", Fourth Edition, McGraw Hill, 2017.
3. D. A. Lind, W. G. Marchal and S. A. Wathen, "Statistical Techniques in Business and Economics", Thirteenth Edition, Tata McGraw Hill, 2007.

AI531H Applications of AI & its impact on Society

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
3	AI531H	Applications of AI & its impact on Society	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Applications of Artificial Intelligence in medicine and health care, heavy industry, transportation and autonomous vehicles, finance, business, security, telecommunications, AR/VR, smart home, smart cities etc. Computer Science Applications - computer vision, pattern recognition, natural language processing, knowledge representation, automated reasoning, bio-computing, semantic web, game theory, data mining, Future of AI and its influence on the future world, AI - Risks and benefits, Regulatory Framework for AI, Artificial General Intelligence

Text Books:

1. Russell, Norvig, "Artificial Intelligence: A Modern Approach", Third Edition, Prentice Hall, 2010.

AI531I Soft Computing

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
1	AI531I	Soft Computing	Elective	4	3	0	2	Yes	Yes	15	20	40	25	0

Soft Computing & Artificial Intelligence: Soft Computing Introduction, Hard Computing, Types of Soft Computing Techniques, Applications of Soft Computing, AI Search Algorithm, Predicate Calculus, Rules of Inference, Semantic Networks, Frames, Objects, Hybrid Models

Artificial Neural Networks : History, overview of biological Neuro-system, Neuron Model, Neural Network Architecture, Learning Rules, Perceptrons, Single Layer Perceptrons, Multilayer Perceptrons, Back propagation Networks: Kohonen's self organizing networks, Hopfield network, Applications of NN

Fuzzy Logic and Fuzzy Sets : Introduction to Fuzzy Logic, Classical and Fuzzy Sets: Complement, Intersections, Unions, Combinations of Operations, Aggregation Operations, Overview of Classical Sets, Membership Function, Fuzzy rule generation

Fuzzy Arithmetic: Fuzzy Numbers, Linguistic Variables, Arithmetic Operations on Intervals & Numbers, Lattice of Fuzzy Numbers, Fuzzy Equations.

Neuro - Fuzzy Modeling : Neuro Fuzzy Controls ,Adaptive Networks Based Fuzzy Interface Systems, Classification and Regression Trees, Data Clustering Algorithms, Rule Based Structure Identification,

Evolutionary Computation

Genetic Algorithms and Swarm Optimizations: Introduction to Genetic Algorithm, Fitness Computations, Evolutionary Programming, Genetic Programming Parse Trees, Variants of GA, Applications, Ant Colony Optimization, Particle Swarm Optimization, Artificial Bee Colony Optimization.

Text Books:

1. Anderson J.A., "An Introduction to Neural Network", PHI, 1995.
2. S.Patnaik, B.Zhong, "Soft Computing Techniques in Engineering Applications", Springer 2014.

Reference Books:

1.	Hertz J. Krogh, R.G. Palmer, Introduction to the Theory of Neural Computation, Addison-Wesley,	1991.
2.	Klir, G. J and Yuan B.B, “Fuzzy Sets and Fuzzy Logic”, Pearson Education India, 2015.	
3.	M. Mitchell, An Introduction to Genetic Algorithm, PHI, 1998.	
4.	S. Kaushik, Artificial Intelligence, Cengage Learning, 2007.	

AI531J Advance Computer Graphics

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
3	AI531A	Advance Computer Graphics	Elective	4	3	0	2	Yes	Yes	15	20	40	25	0

Line Drawing and transformation: Basic raster graphical algorithm for 2D primitives, Line drawing algorithm, 2D and 3D transformation. **Clipping:** Window, View port, clipping algorithm. **Curves and Surfaces:** Circle drawing algorithm, Ellipse drawing algorithm, Bezier curve, bspline curve, surfaces, Solid modelling. **Projection:** Parallel projection, Perspective projection, Computation of vanishing point. **Visible surface determination:** Z-buffer algorithm, Scan line algorithm, Area subdivision algorithm, Raytracing algorithm. **Shading:** Illumination mode, Specular reflection model, Shading models for curve surfaces, Radiosity method, Rendering, Recursive ray tracing, Texture mapping Advanced Modelling Techniques Procedural Models, Fractal Models, Grammar based models, particle systems. **Animation:** 3D animation, morphing, simulation of key frames

Text Books:	
1.	Foley, “Computer Graphics Principles & Practice”, Second Edition, Pearson Education.
2.	Hearn & Baker, “Computer Graphics C version”, Second Edition, Pearson Education.
3.	Roger and Adams, “Mathematical Element for Computer Graphics”, Second Edition, Tata McGraw Hill.
4.	David F. Rogers, “Procedural Element for computer graphics”, McGraw Hill Book Company.

AI531K Robotics Engineering

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
3	AI531K	Robotics Engineering	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Robotics Engineering: History of robots, Classification of robots, Present status and future trends, Basic components of robotic system, Basic terminology- Accuracy, Repeatability, Resolution, Degree of freedom, Mechanisms and transmission, End effectors: Grippers-different methods of gripping, Mechanical and other types of grippers, Tools as end effectors, Robot and effector interface, Gripper selection and design, Specifications of robot. **Drive systems and Sensors:** Drive system- hydraulic, pneumatic and electric systems Sensors in robot – Touch sensors, Tactile sensor, Proximity and range sensors, Robotic vision sensor, Force sensor, Light sensors, Pressure sensors. **Kinematics and Dynamics of Robots:** 2D, 3D Transformation, Scaling, Rotation, Translation, Homogeneous coordinates, multiple transformation, Simple problem, Matrix representation, Forward and Reverse Kinematics Of Three Degree of Freedom, Homogeneous Transformations, Inverse kinematics of Robot, Robot Arm dynamics, D-H representation of robots, Basics of Trajectory Planning, Robot Dynamics: Lagrangian Mechanics, Lagrangian Formulation and numericals, Dynamics, Newton-Euler Recursive Algorithm, Simulation, Euler-Lagrange Equations of motion/Any one other formulation like using Decoupled Natural Orthogonal Complements (DeNOC). **Robot Controls and Programming:** Robot controls-Point to point control, Continuous path control, Intelligent robot, Control system for robot joint, Control actions, Feedback devices, Encoder, Resolver, LVDT, Motion Interpolations, Adaptive control, Introduction to Robotic Programming, On-line and offline programming, programming examples. **Applications for Manufacturing:** Flexible automation, Robot cell layouts, Machine interference, Other considerations in work cell design, Work cell control, interlocks, Robot cycle time analysis, Mechanical design of robot links, Typical applications of robots in material transfer, machine loading/unloading; processing operations; assembly and inspection.

Text Books:

1.	M. P Groover, N.G Odrey, M. Weiss, R.N Nagel, "Technology programming and Applications", McGraw Hill, 2012.
2.	Craig. J. J., "Introduction to Robotics- mechanics and control", Addison- Wesley, 1999.
3.	Saha, S.K., "Introduction to Robotics, Second Edition, McGraw-Hill Education", New Delhi, 2014.
Reference Books:	
1.	S.R. Deb, "Robotics Technology and flexible automation", Tata McGraw-Hill Education., 2009.
2.	R. D. Klafter, Thomas. A, C.Elewski, M. Negin, "Robotics Engineering an Integrated Approach", PHI Learning., 2009
3.	R. Asfahl. C, "Robots and Manufacturing Automation", John Wiley & Sons Inc., 1985.
4.	Fu. K. S., Gonzalez. R. C. & Lee C.S.G, "Robotics control, sensing, vision and intelligence", McGraw Hill Book co, 1987.

AI532A Speech and Biometric Processing

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532A	Speech and Biometric Processing	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Basic Text Processing: Tokenization, Stemming, Language Modeling: N-grams, smoothing, Morphology, Parts of Speech Tagging, Syntax: PCFGs, Dependency Parsing, Distributional Semantics, Lexical Semantics, Word Sense Disambiguation, Information Extraction: Relation extraction, Event extraction, Text Summarization, Text Classification, Machine Translation.

Text Books:

1.	Daniel Jurafsky and James H. Martin, “Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics”, Second Edition. Prentice-Hall, 2009.
2.	A. James, “Natural language Understanding “ Second Edition, Pearson Education, 1994.
3.	A. Bharati, R. Sangal, V. Chaitanya, “Natural language processing: a Paninian perspective”, PHI, 2000.

AI532B Data Mining and Data Warehousing

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532B	Data Mining and Data Warehousing	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Data Mining Concepts: Data mining primitives, Basics of data mining, Query language, Architectures of data mining systems. **Mining Association Rules in Large Databases:** Association Rule Mining, Mining Single Dimensional Boolean Association Rules from Transactional Databases, Mining Multilevel Association Rules from Transaction Databases, Mining Multidimensional Association Rules from Relational Databases and Data Warehouses, From Association Mining to Correlation Analysis, Constraint Based Association Mining. **Classification and Prediction:** Issues Regarding Classification and Prediction, Classification by Decision Tree Induction, Bayesian Classification, Classification by Back propagation, Classification Based on Concepts from Association Rule Mining, Other Classification Methods, Prediction, Classifier Accuracy. **Cluster Analysis in Data Mining:** Types of Data in Cluster Analysis. A Categorization of Major Clustering Methods, Partitioning Methods, Density Based Methods, Grid Based Methods; Model Based Clustering Methods, Outlier Analysis. **Mining Complex Types of Data:** Multidimensional Analysis and Descriptive Mining of Complex Data Objects, Mining Spatial Databases, Mining Multimedia Databases, Mining Time Series and Sequence Data, Mining Text Databases. **Applications and trends in Data Mining:** - Applications, Systems products and research prototypes, Additional themes in data mining, Trends in Data mining, spatial mining, and Web Mining.

Test Books:

1. P. Ponnian, "Data Warehousing Fundamentals", John Wiley
2. M.H. Dunham, Pearson Education, "Data Mining Introductory & Advanced Topics", 2011.
3. H. Kamber, M. Kaufman, "Data Mining Concepts & Techniques", Elsevier, 2011.
4. R. Kimball, The "Data Warehouse Lifecycle Tool Kit", John Wiley, 2012.

5. M. Berry, G. Linoff, "Master in Data Mining", John Wiley, 1996.

AI532C Deep Learning

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532C	Deep Learning	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Pre- Requisite: Linear Algebra, Probability and Statistics, Multivariate Calculus, Basics of Machine Learning.

Details of Course

Introduction to Deep Learning and Neural Networks

Overview and intuition of deep learning, applications of deep learning, mathematical preliminaries (linear algebra, probability, calculus), neural network architecture (neurons, weights, layers), hidden units and activation functions (sigmoid, tanh, ReLU), empirical loss and loss functions (MSE, cross-entropy), gradient-based learning, introduction to backpropagation.

Optimization and Regularization Techniques

Optimization for training deep models (mini-batch gradient descent), advanced optimization techniques (momentum, RMSprop, Adam), understanding local/global optimum, learning rate strategies (decay schedules, warm restarts), train/dev/test set configuration, bias-variance trade-off, underfitting and overfitting, regularization techniques (L1/L2 regularization, dropout), input normalization, batch and layer normalization.

Convolutional Neural Networks (CNNs)

Introduction to convolutional neural networks (convolution operation, feature maps), CNN architecture and layers (convolution, pooling, fully connected layers), pooling layers and techniques (max pooling, average pooling), deep CNN architectures (LeNet, AlexNet,

VGGNet, ResNet, Inception), data augmentation in CNNs, transfer learning and pre-trained models.

Recurrent Neural Networks (RNNs) and Sequence Modeling

Introduction to sequence models (RNNs, LSTMs, GRUs), vanishing/exploding gradients in RNNs, backpropagation through time (BPTT), bidirectional RNNs, deep RNNs, optimization for long-term dependencies, attention mechanisms for sequences, applications of RNNs in NLP and time-series data.

Transformers and Attention Mechanisms

Fundamentals of transformers (architecture, self-attention), multi-head attention mechanism, positional encoding, sequence embeddings, applications of transformers in NLP (BERT, GPT), introduction to vision transformers (ViT), comparison between RNNs and transformers, advantages of attention over recurrence.

Generative Models and Advanced Applications

Introduction to generative models (latent space, autoencoders), structure and applications of autoencoders, variational autoencoders (VAEs), basics of generative adversarial networks (GANs), training GANs (generator and discriminator roles), advanced GANs (conditional GANs, CycleGAN), use cases of generative models (image generation, style transfer).

S. No.	Author(s)	Book Title	Edition	Year	Publisher
1	Ian Goodfellow, Yoshua Bengio, Aaron Courville	Deep Learning	1st	2016	MIT Press
2	Charu C. Aggarwal	Neural Networks and Deep Learning: A Textbook	1st	2018	Springer
3	Aurélien Géron	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow	2nd	2019	O'Reilly Media

4	Christopher M. Bishop	Pattern Recognition and Machine Learning	1st	2006	Springer
5	François Chollet	Deep Learning with Python	1st	2018	Manning Publications
6	Maxim Lapan	Deep Reinforcement Learning Hands-On	2nd	2020	Packt Publishing
7	David Barber	Bayesian Reasoning and Machine Learning	1st	2012	Cambridge University Press
8	Lewis Tunstall, Leandro von Werra, Thomas Wolf	Natural Language Processing with Transformers	1st	2022	O'Reilly Media
9	Kevin P. Murphy	Probabilistic Machine Learning: An Introduction	1st	2022	MIT Press
10	Michael Nielsen	Neural Networks and Deep Learning (Online Book)	-	2015	- (Freely available online)

AI532D Computer Vision

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532D	Computer Vision	Elective	4	3	0	2	Yes	Yes	15	20	40	25	0

Fundamentals of Computer Vision and Image Processing

Overview of Computer Vision, Image formation and representation, Understanding pixels and image types (RGB, grayscale, binary), Image sensors and camera models, Mathematical foundations (Linear Algebra, Probability, Statistics), Image enhancement techniques (smoothing, sharpening, filtering), Edge detection (Sobel, Canny), Image transformations (scaling, rotation, affine transformations), Histogram equalization, Contour detection, Feature extraction (SIFT, SURF, ORB, HOG).

Image Classification and Deep Learning Fundamentals

Image classification challenges and tasks, Traditional classification approaches (K-Nearest Neighbors, Support Vector Machines, Bag of Visual Words), Introduction to Neural Networks, Deep learning for image classification, Convolutional Neural Networks (CNNs), Key architectures (LeNet, AlexNet, VGG, ResNet, EfficientNet), Transfer learning and fine-tuning, Evaluation metrics for classification (accuracy, precision, recall).

Object Detection and Image Segmentation Techniques

Introduction to object detection, Challenges in localization and recognition, Classical methods (Sliding window, Template matching, Viola-Jones face detection), Deep learning for object detection (R-CNN, Fast R-CNN, Faster R-CNN, YOLO, SSD), Anchor boxes and multi-scale detection, Introduction to image segmentation, Semantic vs. instance segmentation, Deep learning for segmentation (Fully Convolutional Networks, U-Net, Mask R-CNN), Advanced CNNs (Inception networks, MobileNet), Metrics for detection and segmentation (IoU, mAP)

Video Representation, Processing, and Temporal Analysis

Difference between image and video processing, Video representation (frame extraction, spatio-temporal information), Frame rates, resolution, and codecs, Temporal coherence in videos, Optical flow computation (Lucas-Kanade, Horn-Schunck), Spatio-temporal representations (3D-CNNs, two-stream networks), Temporal pyramids, Real-time object detection in videos (YOLO, SSD), Video object tracking (Mean-shift, Kalman Filter, correlation filters, GOTURN), Handling occlusions, Re-identification in video tracking

Action Recognition and Video-based Applications

Introduction to action recognition, Challenges in understanding actions from sequences, Feature-based techniques for action recognition (OpenPose, HOG, Motion History Images), Deep learning for action recognition (RNNs, LSTMs for sequence modeling, 3D-CNNs), Advanced methods (Two-stream networks for RGB and optical flow, Temporal Convolutional Networks), Attention mechanisms in temporal vision, Applications in human activity detection, Temporal modeling for video understanding.

3D Vision, Applications, and Emerging Topics

Camera models and calibration (pinhole camera model, intrinsic and extrinsic parameters), 3D reconstruction and depth estimation (stereo vision, epipolar geometry, structure from motion), Point cloud processing and 3D object recognition, Applications in autonomous vehicles (SLAM), Face detection and recognition (DeepFace, FaceNet), Scene understanding (scene classification, semantic labeling), Medical image analysis (segmentation, diagnostics), Generative models in vision (GANs, VAEs for image/video synthesis), Ethics and fairness in computer vision models, Vision applications in robotics and automation.

1. **Richard Szeliski**, *Computer Vision: Algorithms and Applications*, 1st Edition, Springer, 2010.
2. **Gonzalez, Rafael C., and Richard E. Woods**, *Digital Image Processing*, 4th Edition, Pearson, 2018.
3. **David A. Forsyth, Jean Ponce**, *Computer Vision: A Modern Approach*, 2nd Edition, Pearson, 2012.
4. **Ian Goodfellow, Yoshua Bengio, Aaron Courville**, *Deep Learning*, 1st Edition, MIT Press, 2016.
5. **Adrian Rosebrock**, *Deep Learning for Computer Vision: Image Classification, Object Detection, and Face Recognition in Python*, 1st Edition, PyImageSearch, 2019.
6. **Simon J.D. Prince**, *Computer Vision: Models, Learning, and Inference*, 1st Edition, Cambridge University Press, 2012.
7. **Bishop, Christopher M.**, *Pattern Recognition and Machine Learning*, 1st Edition, Springer, 2006.
8. **Hartley, Richard, and Andrew Zisserman**, *Multiple View Geometry in Computer Vision*, 2nd Edition, Cambridge University Press, 2003

AI532E Semantic Web

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532E	Semantic Web	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Semantic Web Vision: Today's web, Examples of semantic web from today's web, Semantic web technologies, layered approach. **Structured web documents in XML:** The XML language, Structuring, Namespaces, Querying and Addressing XML documents, Processing. **Describing Web Resources:** Introduction, RDF, RDF Schema syntax and language, Direct Inference System, Querying RQL. **Web Ontology Language:** Introduction, OWL language, Examples, OWL in OWL, Future extensions. **Logic and Inference:** Rules: Introduction, Monotonic Rules syntax, semantics & examples, Nonmonotonic rules – syntax & examples, Encoding in XML. **Ontology Engineering:** Introduction, Manual construction of Ontology, Reusing existing ontology, using Semi-automatic methods, Knowledge semantic web architecture. **Applications & Future trends:** Various applications, Issues and future trends.

Text Books:

1. P. Hitzler, M. Krötzsch, "Foundations of Semantic Web Technologies", CRC Press, 2009.
2. G. Antoniou and F. Hermelen, "A Semantic web Primer", MIT Press, 2004.
3. John Hebler, "Semantic Web programming", Wiley, 2009.

AI532F Security and Privacy in Social Networks

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532F	Security and Privacy in Social Networks	Elective	4	3	1	0	Yes	No	25	25	50	0	0

What are Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs. Collecting data from Online Social Media. Trust, credibility, and reputations in social systems. Online social Media and Policing. Information privacy disclosure, revelation and its effects in OSM and online social networks. Phishing in OSM & Identifying fraudulent entities in online social networks.

Text Books:

1. T. Segaran, "Programming Collective Intelligence: Building Smart Web 2.0 Applications", O'reilly Media, 2007.
2. Q. Zervaas, "Practical Web 2.0 Applications with PHP", Third Edition, APRESS, 2008.
3. G. Bell, "Building Social Web Applications", Oreilly, 2009.
4. D. Stuttard and M. Pinto, "The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws", Second Edition, Wiley, 2007.

AI532G Mobile and Cellular Network Security

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532G	Mobile and Cellular Network Security	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction: History and basics of cellular systems, performance criteria: voice quality, data quality, picture/vision quality, service quality, Mobile radio environment, operation of cellular systems, Frequency reuse channels. **Application level security in cellular networks:** Generations of cellular networks, security issues in cellular networks, GSM security for applications, GPRS, EDGE, UMTS security for applications, 3G security for applications, security and authentication. **Mobile HTML security and wireless network security:** Authentication on mobile sites, encryption, application attacks on mobile sites, mobile browser weaknesses, enterprise security on mobile operating system, wireless LAN security. **Mobile malware and testing tools:** Mobile malware and threat scenarios, mobile security penetration testing tools: manifest explorer, intent sniffer, intent fuzzer; browser extensions: WMLBrowser, user agent switcher, FoxyProxy, TamperData; networking tools: wireshark, tcpdump, scapy; web application tools: WebScarab, Gizmo. **Mobile security framework:** Security framework for mobile environment, mobile agents and its application domains, mobile web security, security of mobile VOIP communications, latest trends in mobile security.

Text Books:	
1.	William C. Y. Lee, “Mobile Cellular Telecommunications”, Third Edition, McGraw Hill, 2001.
2.	‘Mobile Application Security”, Himanshu Dwivedi, Chris Clark and David Thiel, First Edition 2010.
3.	“Wireless & Mobile Network Security”: Pallapa Venkataram, Satish Babu, TMH, 2010.
4.	Jochen Schiller, “Mobile Communications”, Addison-Wesley, Second Edition, 2004.
5.	Raj Kamal, “Mobile Computing”, Oxford University Press, 2007.

AI532H Game Theory

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532H	Game Theory	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Game Theory: What is game theory. Theory of rational choice, Interacting decision makers. **Strategic Games and Nash Equilibrium:** Strategic games: examples, Nash equilibrium: concept and examples, Best response functions, Dominated Actions, Symmetric games and symmetric equilibria. **Illustrations of Nash Equilibrium and Extensive Games:** Cournot's model of duopoly market, Bertrand's model of duopoly market, Electoral Competition, War of Attrition, Auctions, Accident Laws. Stackelberg model of duopoly markets, Ultimatum game. **Mixed Strategy Nash Equilibrium:** Introduction, Strategic games with randomisation, Mixed strategy Nash equilibrium: concept and examples, Dominated Actions, Formation of Players' beliefs. **Extensive Games and Nash Equilibrium:** Introduction to extensive games, Strategies and outcomes, Nash equilibrium, Subgame perfect Nash equilibrium, Backward induction. **Research Trends in Game Theory:** Research trends include dynamic games, games of incomplete information, and evolutionary games, applications in the fields of voting, bargaining, lobbying and violent conflict

Text Books:

1. Nisan, Roughgarden, Tardos, Vazirani, "Algorithmic Game Theory", Cambridge University Press, 2007.
2. M. Osborne, "An Introduction to Game Theory", Oxford University Press, 2003.
3. A. MacKenzie, "Game Theory for Wireless Engineers", Synthesis lectures on Communications, 2006.
4. R. Gibbons, "A Primer in Game Theory", Pearson Education, 1992.

AI532I Intelligent Agents

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532I	Intelligent Agents	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction: Overview of artificial intelligence and intelligent agents, agents and environments, **Solving problems by search:** Problem-solving agents, formulation of problems, searching for solutions, uninformed search strategies, breadth first search, depth first search, informed (heuristic) search strategies, greedy best first search, A* search: minimizing the total estimated cost, heuristic functions, local search algorithms and optimization, online search agents and unknown environments, **Adversarial search:** Minimax algorithm, alpha-beta pruning, evaluation functions, cutting off search, **Logical agents:** Knowledge-based agents, wumpus world, logic-based agents, propositional logic: reasoning patterns in propositional logic, resolution, forward and backward chaining, **First order logic:** Syntax and semantics of first-order logic, knowledge engineering in first-order logic, Inference in first order logic: propositional and first-order inference, unification and lifting, forward chaining, backward chaining, resolution, **Uncertainty:** Uncertainty in AI, probability notation, axioms of probability, inference using full joint distributions, independence, Bayes' rule, wumpus world revisited, **Probabilistic reasoning:** Knowledge representation in an uncertain domain, semantics of Bayesian networks, representation of conditional distribution, exact and approximate inference in Bayesian networks, **Statistical learning methods:** Overview, learning with complete data, learning with hidden variables: the EM algorithm.

Text Books:	
1.	Russell, Norvig, "Artificial Intelligence: A Modern Approach", Third Edition, Prentice Hall, 2010.
2.	Nils John Nilsson, "Artificial Intelligence: A New Synthesis", Morgan Kaufmann Publication, 2009.
3.	M. Negnevitsky, "Artificial Intelligence: A Guide to Intelligent Systems", Second Edition, Pearson, 2011.

AI532J Genetic Algorithm & Evolutionary Algorithm

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
2	AI532J	Genetic Algorithm & Evolutionary Algorithm	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Evolutionary Computing: Global Optimization, Components of an evolutionary algorithm, Evolution strategies, Fitness Functions, Learning Classifier systems, Parameter Control, Multi-modal Problems. **Swarm Intelligence:** Introduction to Swarm Intelligence and its application to optimization problems, Particle Swarm Optimization algorithm, position and velocity updation. **Genetic Algorithm:** Genetic algorithm basics: Population and generation of chromosomes, Fitness function, survival of the fittest, reproduction, cross-over and mutation, Genetic algorithm convergence, Genetic programming. **Hybrid Methods and Multi-Objective Evolutionary Algorithms:** Variants of Particle Swarm optimization and Genetic Algorithm, Hybridization of Particle Swarm and Genetic based optimizations, Hybrid Multi-Objective Optimization algorithms. **Recent nature-inspired evolutionary algorithms:** Cuckoo search algorithm, Artificial Bee Colony Optimization, Ant Colony Optimization, Fire-fly algorithm, Bacterial Foraging, Application to the travelling salesman problem. **Application to real world optimization problems:** Optimization examples from Machine Learning, Robotics, Image Processing and Computer Vision, Web and data mining, network traffic routing.

Text Books:	
1.	“Computational Intelligence”, Second Edition, by Andries P. Engelbrech, John Wiley & Sons, 2008.
2.	“An Introduction to Genetic Algorithm”, Melanie Mitchell, MIT Press, 1996.
3.	“Genetic Algorithms in Search, Optimization, and Machine Learning”, David Goldberg, Addison-Wesley, 1989.
4.	“Introduction to Evolutionary Computing”, A.E Eiben and J.E. Smith, Springer, Second Edition, 2007.

5. “Evolutionary Computation, a "Unified Approach"”, K. DeJong, MIT Press, 2006.

AI534A Augmented Reality and Virtual Reality

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534B	Augmented Reality and Virtual Reality	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction of Virtual Reality: Fundamental Concept and Components of Virtual Reality. Primary Features and Present Development on Virtual Reality. Multiple Modals of Input and Output Interface in Virtual Reality: Input -- Tracker, Sensor, Digital Glove, Movement Capture, Video-based Input, 3D Menus & 3DScanner etc. Output -- Visual / Auditory / Haptic Devices Visual Computation in Virtual Reality: Fundamentals of Computer Graphics. Software and Hardware Technology on Stereoscopic Display. Advanced Techniques in CG: Management of Large Scale Environments & Real Time Rendering. Environment Modeling in Virtual Reality: Geometric Modeling, Behavior Simulation, Physically Based Simulation. Interactive Techniques in Virtual Reality: Body Track, Hand Gesture, 3D Manus, Object Grasp. Introduction of Augmented Reality (AR): System Structure of Augmented Reality. Key Technology in AR. Development Tools and Frameworks in Virtual Reality: Frameworks of Software Development Tools in VR. X3D Standard; Vega, MultiGen, Virtools etc. Application of VR in Digital Entertainment: VR Technology in Film & TV Production. VR Technology in Physical Exercises and Games. Demonstration of Digital Entertainment by VR

Text Books:

1. Burdea, G. C. and P. Coffet. “Virtual Reality Technology”, Second Edition. Wiley-IEEE Press, 2003/2006.

Reference Books:

1. Sherman, William R. and Alan B. Craig.” Understanding Virtual Reality – Interface, Application, and Design”, Morgan Kaufmann, 2002.
2. Fei GAO.” Design and Development of Virtual Reality Application System”, Tsinghua Press, March 2012. . Guangran LIU. Virtual Reality Technology, Tsinghua Press, Jan. 2011

AI534B **Big Data Analytics**

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534B	Big Data Analytics	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Big Data: Evolution of Big data, Best Practices for Big data Analytics – Big data characteristics, Validating, The Promotion of the Value of Big Data, Big Data Use Cases- Characteristics of Big Data Applications, Perception and Quantification of Value, Understanding Big Data Storage, A General Overview of High-Performance Architecture, HDFS – MapReduce and YARN, Map Reduce Programming Model.

Clustering and Classification: Advanced Analytical Theory and Methods: Overview of Clustering ,K-means, Use Cases – Overview of the Method, Determining the Number of Clusters, Diagnostics, Reasons to Choose and Cautions, Classification: Decision Trees Overview of a Decision Tree, The General Algorithm, Decision Tree Algorithms Evaluating a Decision Tree, Decision Trees in R, Naïve Bayes, Bayes' Theorem, Naïve Bayes Classifier.

Association and Recommendation System: Advanced Analytical Theory and Methods-Association Rules, Overview – Apriori Algorithm, Evaluation of Candidate Rules, Applications of Association Rules, Finding Association& finding similarity, Recommendation System: Collaborative Recommendation- Content Based Recommendation, Knowledge Based Recommendation, Hybrid Recommendation Approaches.

Classification : Decision Trees, Overview of a Decision Tree, The General Algorithm, Decision Tree Algorithms, Evaluating a Decision Tree, Decision Trees in R, Naïve Bayes, Bayes' Theorem, Naïve Bayes Classifier.

Stream Memory: Introduction to Streams Concepts, Stream Data Model and Architecture, Stream Computing, Sampling Data in a Stream, Filtering Streams, Counting Distinct Elements in a Stream, Estimating moments, Counting oneness in a Window, Decaying Window, Real time Analytics Platform(RTAP) applications, Case Studies, Real Time Sentiment Analysis, Stock Market Predictions. Using Graph Analytics for Big Data: Graph Analytics.

NoSQL Data Management For Big Data And Visualization: NoSQL Databases: Schema-less Models: Increasing Flexibility for Data Manipulation-Key Value Stores- Document Stores, Tabular Stores, Object Data Stores, Graph Databases Hive, Sharding, Hbase, Analyzing big data with twitter, Big data for E-Commerce Big data for blogs, Review of Basic Data Analytic Methods using R.

Text Books:
1. C. Eaton and D. deroos et al., “Understanding Big data ”, McGraw Hill, 2012
2. A. Rajaraman and J.D. Ullman, “Mining of Massive Datasets”, Cambridge University Press, 2012
3. B. Lublinsky, K. T. Smith and A.Yakubovich, “Professional Hadoop Solutions”, Wiley, ISBN: 9788126551071, 2015.
4. T. White, “HADOOP: The definitive Guide” , O Reilly 2012.

AI534C Digital Forensics

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534C	Digital Forensics	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction: Understanding the need of Computer Forensics, Definitions, Computer Hardware: Analysis of sources for digital evidence, Digital Media, Hard disk basics, mobile phones. Forensic Tools: Forensic hardware, Hardware write/blockers, Hard drive acquisitions, Processing the scene. Files and File Systems: Windows file systems, Forensic file images, metadata, File signatures. Forensic software: Different software packages, Basic search queries, ASCII, UNICODE, Regular expressions, viewing and managing keywords and cases, Encryption, password protection, Password recovery tools. Physical evidence: fingerprints or other evidence on machines, keyboards. Forensic Reports: Proper report writing, Explaining forensics to the uneducated. Email analysis: IP tracking, Tracking and analysis of emails, Webmail, POP, IMAP. File signature analysis: File signatures, File extensions, Detecting file manipulation. Hash Analysis: Hashing files, Hash libraries. Window Artifacts: My documents, Recycle bin, Installed programs, Windows XP vs. Windows 7.

Text book:

1. Davis, Philipp, and Cowen, Hacking Exposed: "Computer Forensics", McGraw-Hill Education

Reference Books:

1. R. Boddington," Practical Digital Forensics", Packt Publishing.
2. N. Jain, D. Kalbande," Digital Forensic: The Fascinating World of Digital Evidences", Wiley
3. M.J. Britz," Computer Forensics and Cyber Crime: An Introduction", Pearson.
4. J. Marcella, G. Guilloso, "Cyber Forensics: from data to digital intelligence", Wiley

AI534D Medical AI

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534D	Medical AI	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Machine Learning Applications in Medical domain. Introduction to decision making in Medical domain. Introduction to Biomedical text extraction and Robotic surgeries. Introduction to different types of medical data. Introduction to volume visualization in medical data Prerequisites. Calculus and Linear Algebra. Machine Learning: Overview of Prominent Machine Learning and Data Mining methods with example applications in the Medical Domain-Application of Computational Intelligent techniques for Medicine. Clinical Decision Support in Medicine- Automatic Data Mining for the Best Treatment of a Disease Natural Language Processing in Medicine- Intelligent Personal Health Record- Application of Artificial Intelligence in Minimally Invasive Surgery and Artificial Palpation. Medical Image Data and Visual Perception - Acquisition of Medical Image Data- Medical Volume Data in Clinical Practice - Image Analysis for Medical data visualization.

Text Books:

1. Arvin Agah,” Medical Applications of Artificial Intelligence”, CRC Press, First Edition, 2013.
2. Ton J Cleophas, “Machine Learning in Medicine”, Springer, 2015.
3. Bernhard Preim and Dirk Bartz, “Visualization in Medicine: Theory, Algorithms, and Applications, (Morgan Kaufmann Series in Computer Graphics)”, First Edition, Morgan Kaufmann, 2007.

AI534E Data Visualization

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534E	Data Visualization	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Value of Visualization – What is Visualization and Why do it: External representation – Interactivity – Difficulty in Validation. Data Abstraction: Dataset types – Attribute types – Semantics. Task Abstraction – Analyze, Produce, Search, Query. Four levels of validation – Validation approaches – Validation examples. Marks and Channels

Rules of thumb – Arrange tables: Categorical regions – Spatial axis orientation – Spatial layout density. Arrange spatial data: Geometry – Scalar fields – Vector fields – Tensor fields. Arrange networks and trees: Connections, Matrix views – Containment. Map color: Color theory, Color maps and other channels.

Manipulate view: Change view over time – Select elements – Changing viewpoint – Reducing attributes. Facet into multiple views: Juxtapose and Coordinate views – Partition into views – Static and Dynamic layers – Reduce items and attributes: Filter – Aggregate. Focus and context: Elide – Superimpose – Distort – Case studies.

Text Books / References
1. Tamara Munzner, Visualization Analysis and Design, A K Peters Visualization Series, CRC Press, 2014.
2. Scott Murray, Interactive Data Visualization for the Web, O'Reilly, 2013.
3. Alberto Cairo, The Functional Art: An Introduction to Information Graphics and Visualization, New Riders, 2012

4. Nathan Yau, *Visualize This: The FlowingData Guide to Design, Visualization and Statistics*, John Wiley & Sons, 2011.

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534F	Statistical Machine Learning	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Statistical Theory: Maximum likelihood, Bayes, minimax, parametric versus nonparametric methods, Bayesian versus Non-Bayesian approaches, classification, regression, density estimation. **Convexity and Optimization:** Convexity, conjugate functions, unconstrained and constrained optimization, KKT conditions. **Parametric Methods:** Linear regression, model selection, generalized linear models, mixture models, classification, graphical models, structured prediction, hidden Markov models. **Sparsity:** High dimensional data and the role of sparsity, basis pursuit and the lasso revisited, sparsity, consistency, persistency, greedy algorithms for sparse linear regression, sparsity in nonparametric regression. sparsity in graphical models, compressed sensing. **Nonparametric Methods:** Nonparametric regression and density estimation, nonparametric classification, clustering and dimension reduction, manifold methods, spectral methods, the bootstrap and subsampling, nonparametric Bayes. **Advanced Theory:** Concentration of measure, covering numbers, learning theory, risk minimization, Tsybakov noise conditions, minimax rates for classification and regression, surrogate loss functions. **Kernel Methods:** Mercer kernels, kernel classification, kernel PCA, kernel tests of independence. **Computation:** The EM Algorithm, simulation, variational methods, regularization path algorithms, graph algorithms. **Other Learning Methods:** Semi-supervised learning, reinforcement learning, minimum description length, online learning, the PAC model, active learning.

Text Books:

1. Adler, J. R in a Nutshell: "A Desktop Quick Reference". O'Reilly Media, 2010.
2. Bishop, C. "Pattern Recognition and Machine Learning". Springer-Verlag, 2006.
3. Murphy, K. "Machine Learning: a Probabilistic Perspective". MIT Press, 2012.
4. Witten, I. H., Frank, E. and Hall, M. A. "Data Mining: Practical Machine Learning Tools and Techniques". Morgan Kaufman, 2011.

AI534G Block Chain and Applications

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534G	Block Chain and Applications	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Blockchain, Centralised, Decentralised and Distributed Network, Need for Distributed Record Keeping, Blocks in Blockchain, Hashing Algorithm, Requirements of Hashing Algorithm, Immutable Ledger, Distributed Peer to Peer Network, Blockchain Mining, How blockchain works, Properties of Blockchain, Types of Blockchains, Applications of Blockchain, Byzantine General's Problem, Byzantine Fault Tolerance, Modeling faults and adversaries Byzantine Generals problem, Digital Signature, Consensus algorithms, Competing Chain Problem, Orphan Blocks,

Introduction to Bitcoin, Features of Bitcoin, Bitcoin's Monetary Policy, The Halving Principle, Block Frequency, How mining works?, Nonce, Target, CPU vs GPU Vs ASICs, Mining Pools, Hash Range, Nonce Range, Time Stamp, Mempool, Factors affecting the success of Transaction, Transaction and UTXO, Transaction Fee, Cryptocurrency Wallets, Types of Cryptocurrency Wallets, How Public and Private key works, Segregated Witness, Hierarchical Deterministic Wallet.

Introduction to Ethereum, Ethereum Nodes, Types of Nodes in Ethereum, Full Node, Light Node, Archive Node, Ethereum Account, Types of Ethereum Account, Externally owned account, Contract Account, Smart Contract, Solidity, Bitcoin script vs Solidity, Decentralised Apps, Ethereum Virtual Machine, Ethereum Gas, Gas Limit, Proof of Work vs. Proof of Stake, Sharding, Benefits of Sharding, Decentralised Autonomous Organisations(DAO), DAO Attack, Hard Fork, Soft Fork, Initial Coin Offerings, Tokens, Alt Coins. Hyper ledger Fabrics, NFTS,

Text Books
1. "Blockchain: Blueprint for a New Economy", by Melanie Swan
2. Blockchain: The blockchain for beginners guide to blockchain technology and leveraging blockchain programming", by Josh Thompsons

3. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", Princeton University Press (July 19, 2016).
Reference Books:
1. Antonopoulos, "Mastering Bitcoin: Unlocking Digital Cryptocurrencies".
2. Satoshi Nakamoto, "Bitcoin: A Peer-to-Peer Electronic Cash System".
3. Gavin Wood, "ETHEREUM: A Secure Decentralized Transaction Ledger," Yellow paper.2014.
4. Nicola Atzei, Massimo Bartoletti, and Tiziana Cimoli, "A survey of attacks on Ethereum smart ontracts".

AI534H Quantum AI

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534H	Quantum AI	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Quantum Artificial Intelligence, Complexity theory, Decision problems, P & NP, Logic based operators, Frames, Categorical representation, Binary vector representation, Production System, Deduction systems, Reaction systems, Conflict resolution, Reversible Computation, Reversible circuits, Fundamentals of Quantum Physics and Quantum Mechanics, Hilbert space , Quantum Time Evolution, Von Neumann Entropy, Heisenberg's uncertainty principle, Randomness, Qubits, Quantum Boolean Circuits, Quantum Fourier Transform, Quantum Oracle, Grover's Amplification, Generate & Test, Quantum Problem Solving, Quantum Tree Search, Quantum Production System, Quantum Computer - Architecture and Representation, Quantum Cognition, Quantum walk on a graph, Quantum annealing, Optimization, Quantum Neural Computation, Quantum annealing, Quantum Computer simulation tool kits.

Text Books:

1. Andreas Wichert, "Principles of Quantum Artificial Intelligence", First Edition, World Scientific Publishing, 2014.
2. Peter Wittek, "Quantum Machine Learning", First Edition, Academic Press, 2014.

AI534I INTELLECTUAL PROPERTY RIGHTS

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534I	INTELLECTUAL PROPERTY RIGHTS	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Scope of Cyber Laws: Nature of Cyber Space, Cyber Property, Cyber Personality, Cyber Transactions. Law of Digital Contracts – Digital Contract – Definition; Formation of Digital. Contracts, System of Digital Signature, Role and Function of Certifying Authorities, the Science of Cryptography. **Nature of Intellectual Property:** Patents, Designs, Trademarks and Copyright. Process of Patenting and Development: technological research, innovation, patenting, development. **Patent Rights:** Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications Information Technology Act, 2000, International Scenario in Cyber Laws, IPR Policies. WIPO, National IPR Policy. New Developments in IPR Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR and IITs. Product Design Importance of product design in industry. Principal requirements of good product design. Factors and considerations affecting product design. Ergonomic factor in product design. Product design methodology and techniques. Basic elements and concepts of visual design. **Product Design Standards:** Standards related to Materials, forms, functions, color, graphics, product development and testing. Packaging materials their characteristics and applications. Packaging design considerations.

Text Books:

1. "Introduction to intellectual property: Theory and Practice", By World Intellectual Property Organization.
2. "Intellectual Property Rights: Innovation, Governance and the Institutional Environment", Birgitte Andersen, Edward Elgar Publishing Limited.
3. "Innovation, Intellectual Property, and Economic Growth" By Christine Greenhalgh, Mark Rogers.

Reference Books:

1. “Intellectual Property Rights, Development, and Catch-Up”, By Hiroyuki Odagiri and Atsushi Sunami.
2. Schechter, Roger E., and John R. Thomas. Intellectual Property: The Law of Copyrights, Patents and Trademarks. New York: West/Wadsworth, 2003.
3. “The global challenge of intellectual property rights” Robert C. Bird, Subhash Chander Jain.
4. “Digital Media and Intellectual Property” By Nicola Lucchi.
5. “The international political economy of intellectual property rights” By Meir Perez Pugatch.

AI534J Internet of Things, Applications and Security

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534J	Internet of Things, Applications and Security	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Internet of Things: Definition and Characteristics of IoT, Physical Design of IoT – IoT Protocols, IoT communication models, IoT Communication APIs IoT-enabled Technologies – Wireless Sensor Networks, Cloud Computing, Big data analytics, Communication protocols, Embedded Systems, IoT Levels and Templates Domain Specific IoTs – Home, City, Environment, Energy, Retail, Logistics, Agriculture, Industry, health and Lifestyle. **IoT and M2M:** Software defined networks, network function virtualization, difference between SDN and NFV for IoT Basics of IoT System Management with NETCONF, YANG- NETCONF, YANG, SNMP NETOPeer. **Introduction to Python:** Language features of Python, Data types, data structures, Control of flow, functions, modules, packaging, file handling, data/time operations, classes, Exception handling Python packages – JSON, XML, HTTPLib, URLLib, SMTPLib. **IoT Physical Devices and Endpoints:** Introduction to Raspberry PI-Interfaces (serial, SPI, I2C) Programming – Python program with Raspberry PI with focus of interfacing external gadgets, controlling output, and reading input from pins. **IoT Physical Servers and Cloud Offerings:** Introduction to Cloud Storage models and communication APIs Webserver – Web server for IoT, Cloud for IoT, Python web application framework designing a RESTful web API.

Text Books:

1. A. Bahga and V. Madisetti, “Internet of Things – A Hands-on Approach”, Universities Press, 2015.
2. M. Richardson and S. Wallace, Getting Started with Raspberry Pi, O’Reilly (SPD), 2014.
3. J. Holler, V. Tsiatsis, C. Mulligan, S. Avesand, S. Karnouskos and D. Boyle, “From Machine-toMachine to the Internet of Things: Introduction to a New Age of Intelligence”, First Edition, Academic Press, 2014.

4.	B.S. Reiter and F.Michahelles, “Architecting the Internet of Things”, Springer, 2011.
5.	W. Stallings, “Foundations of modern networking: SDN, NFV, QOE, IOT, and cloud” publisher: Addison-Wesley, 2015.

AI534K Business Data Analytics

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534K	Business Data Analytics	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction to Big Data: Evolution of Big data, Best Practices for Big data Analytics – Big data characteristics, Validating, The Promotion of the Value of Big Data, Big Data Use Cases- Characteristics of Big Data Applications, Perception and Quantification of Value, Understanding Big Data Storage, A General Overview of High-Performance Architecture, HDFS – MapReduce and YARN, Map Reduce Programming Model. **Clustering and Classification:** Advanced Analytical Theory and Methods: Overview of Clustering ,K-means, Use Cases – Overview of the Method, Determining the Number of Clusters, Diagnostics, Reasons to Choose and Cautions, Classification: Decision Trees Overview of a Decision Tree, The General Algorithm, Decision Tree Algorithms Evaluating a Decision Tree, Decision Trees in R, Naïve Bayes, Bayes’ Theorem, Naïve Bayes Classifier. **Association and Recommendation System:** Advanced Analytical Theory and Methods- Association Rules, Overview – Apriori Algorithm, Evaluation of Candidate Rules, Applications of Association Rules, Finding Association& finding similarity, Recommendation System: Collaborative Recommendation- Content Based Recommendation, Knowledge Based Recommendation, Hybrid Recommendation Approaches. **Classification :** Classification: Decision Trees, Overview of a Decision Tree, The General Algorithm, Decision Tree Algorithms, Evaluating a Decision Tree, Decision Trees in R, Naïve Bayes, Bayes’ Theorem, Naïve Bayes Classifier. **Stream Memory:** Introduction to Streams Concepts, Stream Data Model and Architecture, Stream Computing, Sampling Data in a Stream, Filtering Streams, Counting Distinct Elements in a Stream, Estimating moments, Counting oneness in a Window, Decaying Window, Real time Analytics Platform(RTAP) applications, Case Studies, Real Time Sentiment Analysis, Stock Market Predictions. Using Graph Analytics for Big Data: Graph Analytics. **NoSQL Data Management For Big Data And Visualization:** NoSQL Databases: Schema-less Models: Increasing Flexibility for Data Manipulation-Key Value Stores- Document Stores, Tabular Stores, Object Data Stores, Graph Databases Hive, Sharding, Hbase, Analyzing big data with twitter, Big data for E-Commerce Big data for blogs, Review of Basic Data Analytic Methods using R.

Text Books:

1.	C. Eaton and D. deroos et al., “Understanding Big data ”, McGraw Hill, 2012
2.	A. Rajaraman and J.D. Ullman, “Mining of Massive Datasets”, Cambridge University Press, 2012
3.	B. Lublinsky, K. T. Smith and A.Yakubovich, “Professional Hadoop Solutions”, Wiley, ISBN: 9788126551071, 2015.
4.	T. White, “HADOOP: The definitive Guide” , O Reilly 2012.

AI534L Predictive Analytics and Statistics

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CWS	MTE	ETE	PRS	PRE
4	AI534L	Predictive Analytics and Statistics	Elective	4	3	1	0	Yes	No	25	25	50	0	0

Introduction: classification & prediction, key ingredients of predictive models, goals of a regression analysis, regression models, data in a regression analysis. **Data preparation:** analyzing the metric data, outlier analysis, correlation analysis, attribute reduction methods, attribute extraction. **Statistical tests:** categories, one-tail and two-tail, Type I and Type II errors, interpreting significance results. **Model Development:** data partition, attribute reduction, model construction, model validation, hypothesis testing, results interpretation, cross-validation. Hypothesis testing and model-comparison tests. **Model evaluation:** performance measures for categorical and continuous dependent variables, ROC analysis.

Text Books:

1. M. Kuhn and K. Johnson, “Applied Predictive Modelling”, Springer Verlag, 2013.
2. R. Malhotra, “Empirical Research in Software Engineering: Concepts, Analysis & Applications”, CRC press, 2016.
3. E.E. Frees, E.W. Derrig, and G. Meyers, “Predictive Modeling Techniques in Actuarial Science”, Vol. I: Predictive Modeling Techniques. Cambridge University Press, 2014.

UCC502 Research Methodology and IPR

Teaching and Examination Scheme				Credits	Hours/Week			Exam Duration (Hrs)		Relative Weights (%)				
Sem	Subject Code	Course Title	Type		L	T	P	TH	PR	CW S	MTE	ETE	PRS	PRE
2	UCC502	Research Methodology and IPR		4	3	1	0	Yes	No	25	25	50	0	0

Pre-requisites: Basic understanding of fundamentals of quantitative techniques

Unit -I:

Introduction, Concepts of research, Meaning and objectives of research, Research process, Types of research, Research problem identification, Research proposal-contents, Funding agencies, Ethical aspects and Plagiarism detection tools.

Unit – II:

Research design process, Need and types of research design, Literature survey, Use of internet and advanced search techniques, Various reputed publishers and their databases, Identification of research gaps, Measurement and scaling techniques, Data collection-types and methods, Processing and analysis of data, Design and analysis of experiment.

Unit – III:

Quantitative techniques, sampling fundamentals, Type of hypothesis, Introduction and applications of Binomial, Normal and Poisson distributions, Statistical tests: Chi-squared test, t-test, f-test etc., Multivariate analysis, Introduction to various statistical analysis software.

Unit – IV:

Computer applications in research, Pre-writing considerations, Principles of thesis and report writing, Formats for thesis, report and research papers, Documentation and presentation tools- introduction to LATEX and MS Office.

Unit-V:

Nature of Intellectual Property: Patents, Designs, Trade and Copyright, Process of Patenting and Development: technological research, innovation, patenting, development, New developments in IPR: National and international scenarios.

Text Books:	
1.	Stuart Melville and Wayne Goddard, “Research Methodology: An Introduction for Science & Engineering Students”, Juta and Company Ltd.

2.	Dr. C.R. Kothari, “Research Methodology: Methods and Techniques”, New Age International Publisher.
3.	Wadehra B.L., “Law Relating to Patents, Trademarks, Copyright Designs and Geographical Indicators”, Universal Law Publishing.

Reference Books:

1.	Ranjit Kumar, “Research Methodology: A Step by Step Guide for Beginners”, 2nd Edition, SAGE Publications Ltd.
2.	G.C. Ramamurthy, “Research Methodology”, Dream Tech Press, New Delhi.